Data-Driven Optimization

선형모형 모델링 - 고급

과제 2

과제 2 – 1

1.15 A furniture manufacturer has three plants, which need 500, 700, and 600 tons of lumber weekly. The manufacturer may purchase the lumber from three lumber companies. The first two lumber manufacturers virtually have unlimited supply, and because of other commitments the third manufacturer cannot ship more than 500 tons weekly. The first lumber manufacturer uses rail for transportation and there is no limit on the tonnage that can be shipped to the furniture facilities. On the other hand, the last two lumber companies use trucks that limit the maximum tonnage that can be shipped to any of the furniture companies to 200 tons. The following table gives the transportation cost from the lumber companies to the furniture manufacturers ($ per ton).

LUMBER COMPANY	FURNITURE FACILITY 1	2	3
1	2	3	5
2	2.5	4	4.8
3	3	3.6	3.2

Formulate the problem as a linear program. (We ask the reader to solve this problem in Exercise 8.32.)

- 3개 공장의 목재 수요는 주당 500, 700, 600톤
- 목재는 3개 회사로부터 목재를 구매.
- 공급 제약
  - ✓ 첫 2개 공장은 공급량 제한 없고 3번째 공장은 주당 500톤
- 수송 제약
  - ✓ 첫 번째 목재 회사는 철도수송으로 수송 제한이 없음.
  - ✓ 나머지 2 회사는 트럭 수송으로 200톤까지 수송가능
- 우측은 각 목재회사에서 공장으로의 톤당 수송비용

과제 2 – 2

1.17 A ten-acre slum in New York City is to be cleared. The officials of the city must decide on the redevelopment plan. Two housing plans are to be considered: low-income housing and middle-income housing. The types of housing can be developed at 20 and 15 units per acre respectively. The unit costs of the low- and middle-income housing are $13,000 and $18,000. The lower and upper limits set by the officials on the number of low-income housing units are 60 and 100. Similarly, the number of middle-income housing units must lie between 30 and 70. The combined maximum housing market potential is estimated to be 150 (which is less than the sum of the individual market limits due to the overlap between the two markets). The total mortgage committed to the renewal plan is not to exceed $2 million. Finally, it was suggested by the architectural adviser that the number of low-income housing units be at least 50 units greater than one-half the number of middle-income housing units.

a. Formulate the minimal cost renewal planning problem as a linear program and solve it graphically.

b. Resolve the problem if the objective is to maximize the number of houses being constructed.

- 10 에이커의 슬럼을 재개발하고자 함.
- 에이커 당 저소득 주택 20개, 중간소득 주택 15개 건설 가능
- 주택 건설 비용 저소득 주택 $13,000/중간소득 주택 $18,000
- 건설 guideline
  - 저소득 주택 60개에서 100개 사이
  - 중간소득 주택 30개에서 70개 사이
- 결합 시장 수요 = 150 개
- 재개발 예산: 최대 $2,000,000
- 건축가 제안
  - 저소득 주택 수는 중간소득 주택 수의 절반 보다 50 개 이상 커야 한다.
- 두 문제 풀기: 비용 최소화, 주택수 최대화